

Foundations of Quantum Computing

Week 4: Many-qubit procedures: examples and intuitions

Explore – Study

The main point of quantum technologies is to somehow try to do things which would be more difficult or impossible with classical information technologies. This requires that we do things that are not possible with classical information – and, in particular, which might not even resemble something that it would make sense to do with classical information.

This week, we will take a first look at the ways that we can use quantum information in a manner in which we cannot use classical information. That is, in an attempt to build some intuitions that can be used to understand Grover’s algorithm and Shor’s algorithm. In this week’s Study section you will cover the following topics:

4.1 Quantum teleportation

4.2 Hadamard operations and phase estimation

This Study session uses the QAAL programming language outlined in the reading for this week’s ‘Prepare’ stage. You should skim those notes before reading the Study material that follows and come back to them if you get confused about a particular way that something is described in the Study materials (for this week or for later weeks).

4.1 Quantum teleportation

The first procedure that we consider is not an algorithm, per se, but rather a famous quantum communication protocol: **quantum teleportation**. Note that this does not represent a way of moving objects from one place to another without moving through the space in between. Instead, it is a way to move **information** from one place to another, without moving through the space in between.

4.1.1 The set-up

As in many communication tasks, it is helpful to describe teleportation as involving two different people: in this case, ‘Alice’ and ‘Bob’. In this scenario, Alice wants to send Bob a qubit of data that she has in her possession. However, Bob is far away, and Alice doesn’t have a way to do this both quickly and reliably by just sending the physical system which stores that qubit. Fortunately for both of them, Alice and Bob have planned ahead:

- They have shared two qubits ‘a’ and ‘b’, which together are in a carefully prepared entangled state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}|0,0\rangle + \frac{1}{\sqrt{2}}|1,1\rangle$. They have both managed to keep these two qubits safe from disturbances, so ‘a’ and ‘b’ are still in that joint state.
- They are also able to communicate using classical information (with radio signals, phones or some other ‘normal’ method of communication).

To share the state $|\Phi^+\rangle$, the qubits ‘a’ and ‘b’ would have to have been in the same place to begin with. For instance, this would have been the case if Alice and Bob had been together to prepare; if Alice had both qubits to begin with and sent the qubit ‘b’ in advance of knowing what she would want to transmit to Bob; or if some third person ‘Charlotte’ prepared the two qubits and distributed them to Alice and Bob. In any case, the state $|\Phi^+\rangle$ of ‘a’ and ‘b’ is easy enough to prepare from an initial state of $|0,0\rangle$, using the following QAAL procedure:

```
define Prepare_PhiPlus a, b :
  # prepare the two qubits in the state  $|\Phi^+\rangle = \frac{1}{\sqrt{2}}|0,0\rangle + \frac{1}{\sqrt{2}}|1,1\rangle$ ,
  # assuming an initial state of  $|0,0\rangle$ 
  operand qubit a, b
  H a
  CX a, b
  stop
```

We can verify this easily enough (using subscripts to represent the order of the qubits ‘a’ and ‘b’ in the state-vectors):

$$|0,0\rangle_{a,b} \xrightarrow{H\ a} \frac{1}{\sqrt{2}}|0,0\rangle_{a,b} + \frac{1}{\sqrt{2}}|1,0\rangle_{a,b} \xrightarrow{CX\ a,b} \frac{1}{\sqrt{2}}|0,0\rangle_{a,b} + \frac{1}{\sqrt{2}}|1,1\rangle_{a,b} . \quad (4.1)$$

The teleportation protocol supposes that Alice and Bob already have a shared entangled state of this sort. (Teleportation uses this entangled state, but creating that state is not itself one of the steps of the process).

4.1.2 The teleportation protocol

The state $|\Phi^+\rangle$ is actually one of four orthogonal states on two qubits, known as the **Bell states**:

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{\sqrt{2}}|0,0\rangle + \frac{1}{\sqrt{2}}|1,1\rangle , & |\Psi^+\rangle &= \frac{1}{\sqrt{2}}|0,1\rangle + \frac{1}{\sqrt{2}}|1,0\rangle , \\ |\Phi^-\rangle &= \frac{1}{\sqrt{2}}|0,0\rangle - \frac{1}{\sqrt{2}}|1,1\rangle , & |\Psi^-\rangle &= \frac{1}{\sqrt{2}}|0,1\rangle - \frac{1}{\sqrt{2}}|1,0\rangle . \end{aligned} \quad (4.2)$$

(It is easy to verify that these are orthogonal, for example, that $\langle\Phi^-|\Psi^+\rangle = 0$, by direct calculation). Together, we refer to these states as the **Bell basis**, as we may decompose any two-qubit state $|\psi\rangle$ as a superposition of these states:

$$\begin{aligned} |\psi\rangle &= u_{00}|00\rangle + u_{01}|01\rangle + u_{10}|10\rangle + u_{11}|11\rangle \\ &= \left(\frac{u_{00}+u_{11}}{\sqrt{2}}\right)|\Phi^+\rangle + \left(\frac{u_{01}+u_{10}}{\sqrt{2}}\right)|\Psi^+\rangle + \left(\frac{u_{01}-u_{10}}{\sqrt{2}}\right)|\Psi^-\rangle + \left(\frac{u_{00}-u_{11}}{\sqrt{2}}\right)|\Phi^-\rangle . \end{aligned} \quad (4.3)$$

Figure 4.1: A procedure `MBell` to measure two qubits in the Bell basis. This procedure to measure in the Bell basis works, in effect, by unitarily transforming the Bell states $|\Phi^\pm\rangle$ and $|\Psi^\pm\rangle$ to the computational basis states $|x, z\rangle$; measuring the values of x and z ; and then undoing the unitary transformation to produce a particular Bell state.

```

define MBell p, q -> x, z :
  # perform a Bell basis measurement on qubits p and q,
  # producing an outcome in classical bits x and z as follows:
  #  $|\Phi^+\rangle \mapsto 00$  ;  $|\Phi^-\rangle \mapsto 01$  ;  $|\Psi^+\rangle \mapsto 10$  ;  $|\Psi^-\rangle \mapsto 11$ 
  operand qubit p, q
  operand bit x, z
  CX p, q      # 1. map the Bell-basis states
  H p          # to the computational basis
  Mz q -> x    # 2. measure the pair of qubits
  Mz p -> z    # in the computational basis
  H p          # 3. undo the change of basis,
  CX p, q      # reproducing a Bell-basis state
  stop

```

In principle, Alice could measure a pair of qubits in her position with respect to the Bell basis, collapsing a two-qubit state such as $|\psi\rangle$ preceding to one of the four Bell states, with probabilities given by the squares of the norms of the preceding amplitudes, $\frac{1}{2}|u_{00} \pm u_{11}|^2$ and $\frac{1}{2}|u_{01} \pm u_{10}|^2$.

The way that Alice would probably realise this in practice for some pair of qubits p and q , is by using a procedure such as the one demonstrated in Figure 4.1, which realises a unitary transformation to map these states onto the computational basis and then measure each of the qubits in the $|0\rangle, |1\rangle$ basis.

The way that Alice makes use of a Bell measurement to perform teleportation is as follows. Suppose Alice has some qubit q , containing a single-qubit state $|\alpha\rangle$, which she wants to send to Bob. Recall that a and b are the qubits that Alice and Bob (respectively) possess, which together are in the state $|\Phi^+\rangle$.

1. Alice measures qubits q and a in the Bell basis and sends two bits x and z to Bob.
2. Once Bob receives x and z , he performs the operation ‘ X b ’ if $x = 1$, and the operation ‘ Z b ’ if $z = 1$.

This is described using QAAL instructions in Figure 4.2 (where we suppose that there is some procedure `PrepareAlpha` which prepares q in the state $|\alpha\rangle$). For the sake of brevity, we define operations `ClassicalCtrlX` and `ClassicalCtrlZ` (presented in Figure 4.3), which each take a bit and a qubit as operands and perform an appropriate operation to the qubit, so long as the classical bit is equal to 1.

Figure 4.2: The quantum teleportation protocol. A fragment of a QAAL program, describing quantum teleportation. Note in particular that the qubit **a** only interacts with **b** before the state of **q** is prepared and that the qubit **q** never interacts with the qubit **b**. The subroutines `ClassicalCtrlX` and `ClassicalCtrlZ` are defined in Figure 4.3.

```

# Set-up
# -----
# Prepare shared Bell state:
qubit a, b          # initially in state  $|0,0\rangle$ 
PreparePhiPlus a, b  # prepare a, b in the state  $|\Phi^+\rangle$ 

# ... some time passes ...
qubit q              # A qubit which belongs to Alice
PrepareAlpha q        # Some process which prepares q in a state
 $|\alpha\rangle$ 

# Teleportation protocol
# -----
# 1. Alice's operations:
bit x, z              # Bits to store results of Bell measurement
MBell q, a -> x, z    # Perform the Bell basis measurement
                        # (Alice sends the bits x and z to Bob)

# 2. Bob's operations:
ClassicalCtrlX x, b    # perform X on b, if x = 1
ClassicalCtrlZ z, b    # perform Z on b, if z = 1
stop

```

4.1.3 Analysis

(a) The state of Alice's and Bob's qubits before measurement

Suppose that $|\alpha\rangle = a_0|0\rangle + a_1|1\rangle$. The state of the three qubits **q**, **a** and **b** is then

$$\begin{aligned}
& |\alpha\rangle_q \otimes |\Phi^+\rangle_{a,b} \\
&= a_0|0\rangle_q \otimes \left(\frac{1}{\sqrt{2}}|00\rangle_{a,b} + \frac{1}{\sqrt{2}}|11\rangle_{a,b} \right) + a_1|1\rangle_q \otimes \left(\frac{1}{\sqrt{2}}|00\rangle_{a,b} + \frac{1}{\sqrt{2}}|11\rangle_{a,b} \right) \\
&= \frac{1}{\sqrt{2}}a_0|0,0,0\rangle_{q,a,b} + \frac{1}{\sqrt{2}}a_0|0,1,1\rangle_{q,a,b} + \frac{1}{\sqrt{2}}a_1|1,0,0\rangle_{q,a,b} + \frac{1}{\sqrt{2}}a_1|1,1,1\rangle_{q,a,b} \\
&= \frac{1}{2}a_0\left(|\Phi^+\rangle_{q,a} + |\Phi^-\rangle_{q,a}\right) \otimes |0\rangle_b + \frac{1}{2}a_0\left(|\Psi^+\rangle_{q,a} + |\Psi^-\rangle_{q,a}\right) \otimes |1\rangle_b \\
&\quad + \frac{1}{2}a_1\left(|\Psi^+\rangle_{q,a} - |\Psi^-\rangle_{q,a}\right) \otimes |0\rangle_b + \frac{1}{2}a_1\left(|\Phi^+\rangle_{q,a} - |\Phi^-\rangle_{q,a}\right) \otimes |1\rangle_b . \quad (4.4)
\end{aligned}$$

In the preceding equation, we do nothing but expand the definitions of $|\alpha\rangle$ and $|\Phi^\pm\rangle$ as superpositions on computational basis states, expand the tensor products, and then re-express $|0,0\rangle$, $|0,1\rangle$, $|1,0\rangle$ and $|1,1\rangle$ on qubits **q** and **a** in terms of the Bell basis. No actual transformations of the state have been applied yet, but by expressing the first two

Figure 4.3: Subroutines for classically-controlled unitaries. Two similar subroutines, which use a classical bit to determine whether some unitary operation, in this case either X or Z, is performed on a single-qubit operand.

<pre> define ClassicalCtrlX c, q : operand bit c operand qubit q !jump_if_zero c, @skip X q @skip: stop </pre>	<pre> define ClassicalCtrlZ c, q : operand bit c operand qubit q !jump_if_zero c, @skip Z q @skip: stop </pre>
--	--

qubits in terms of Bell states, we may attempt to quickly describe the effect of the Bell basis measurement. Expanding the preceding equation and collecting the terms for $|\Phi^+\rangle$, $|\Phi^-\rangle$, $|\Psi^+\rangle$ and $|\Psi^-\rangle$, we obtain:

$$\begin{aligned}
& |\alpha\rangle_q \otimes |\Phi^+\rangle_{a,b} \\
&= \frac{1}{2} |\Phi^+\rangle_{q,a} \otimes (a_0 |0\rangle_b + a_1 |1\rangle_b) + \frac{1}{2} |\Phi^-\rangle_{q,a} \otimes (a_0 |0\rangle_b - a_1 |1\rangle_b) \\
&\quad + \frac{1}{2} |\Psi^+\rangle_{q,a} \otimes (a_1 |0\rangle_b + a_0 |1\rangle_b) + \frac{1}{2} |\Psi^-\rangle_{q,a} \otimes (a_1 |0\rangle_b - a_0 |1\rangle_b). \quad (4.5)
\end{aligned}$$

Here, we describe Alice's two qubits as being in a superposition of possible entangled states $|\Phi^\pm\rangle$ and $|\Psi^\pm\rangle$, with Bob's qubit being in some particular (generally different) single-qubit state for each of these states. This does not mean that Alice's two qubits are 'secretly' in some entangled state. In fact, they are completely independent. But the superposition in the preceding equation suggests the way that the state of Bob's qubit will collapse, when Alice performs her Bell-basis measurement.

(b) An observation about Bob's qubit before measurement

Consider the information that is available to Bob at this stage. The only thing that Bob can do is to measure his qubit in some basis $|u\rangle, |v\rangle$. Suppose that $|u\rangle = u_0 |0\rangle + u_1 |1\rangle$, and $|v\rangle = u_1^* |0\rangle - u_0^* |1\rangle$: if Bob were to perform a measurement in this basis at this stage, the probability of obtaining the outcome $|u\rangle$ would be:

$$\begin{aligned}
& \left(\langle \alpha | \otimes \langle \Phi^+ | \right)_{q,a,b} |u\rangle \langle u|_b \left(|\alpha\rangle \otimes |\Phi^+\rangle \right)_{q,a,b} \\
&= \langle \alpha | \alpha \rangle_q \otimes \left(\langle \Phi^+ |_{a,b} |u\rangle \langle u|_b | \Phi^+ \rangle_{a,b} \right) \\
&= \frac{1}{2} \left(\langle 0, 0 |_{a,b} + \langle 1, 1 |_{a,b} \right) |u\rangle \langle u|_b \left(|0, 0\rangle_{a,b} + |1, 1\rangle_{a,b} \right) \\
&= \frac{1}{2} \left(\langle 0 | 0 \rangle_a \otimes \langle 0 | u \rangle \langle u | 0 \rangle_b + \langle 1 | 1 \rangle_a \otimes \langle 1 | u \rangle \langle u | 1 \rangle_b \right) = \frac{1}{2} (u_0^* u_0 + u_1^* u_1) = \frac{1}{2}. \quad (4.6)
\end{aligned}$$

Notice how this answer does not depend on $|\alpha\rangle$. (This makes sense, as Bob may attempt to perform this measurement before Alice even obtains the qubit in state $|\alpha\rangle$. That is, the only information he could obtain is about his half of the entangled state $|\Phi^+\rangle$). What

we observe is that, no matter which basis Bob measures his half of $|\Phi^+\rangle$ in, he obtains a perfectly random outcome. In effect, his qubit contains no useful information whatsoever before Alice performs her measurement.

(c) The effect of Alice's Bell-basis measurement

When Alice performs her measurement, her two qubits collapse into one of the four Bell states $|\Phi^\pm\rangle$ or $|\Psi^\pm\rangle$. From Equation (4.5), we can see that there is a term $\frac{1}{2}|\Phi^\pm\rangle \otimes |\beta\rangle$ or $\frac{1}{2}|\Psi^\pm\rangle \otimes |\beta\rangle$ for each of the Bell states, where $|\beta\rangle$ is some single-qubit state of Bob's qubit (potentially different ones in each of the four cases). The probability of any one of the four outcomes of the Bell-basis measurement is then $|\frac{1}{2}|^2 = \frac{1}{4}$, with the four following possible outcomes:

- $|\Phi^+\rangle_{\mathbf{q},\mathbf{a}} \otimes (a_0|0\rangle + a_1|1\rangle)_{\mathbf{b}}$, if $\mathbf{x} = 0$ and $\mathbf{z} = 0$
- $|\Phi^-\rangle_{\mathbf{q},\mathbf{a}} \otimes (a_0|0\rangle - a_1|1\rangle)_{\mathbf{b}}$, if $\mathbf{x} = 0$ and $\mathbf{z} = 1$
- $|\Psi^+\rangle_{\mathbf{q},\mathbf{a}} \otimes (a_1|0\rangle + a_0|1\rangle)_{\mathbf{b}}$, if $\mathbf{x} = 1$ and $\mathbf{z} = 0$
- $|\Psi^-\rangle_{\mathbf{q},\mathbf{a}} \otimes (a_1|0\rangle - a_0|1\rangle)_{\mathbf{b}}$, if $\mathbf{x} = 1$ and $\mathbf{z} = 1$.

Then Bob's qubit $\mathbf{b}$ is in some single-qubit state and one either identical or very similar to the original state $|\alpha\rangle = a_0|0\rangle + a_1|1\rangle$ of Alice's qubit $\mathbf{q}$. This is despite the fact that $\mathbf{b}$ has never interacted with $\mathbf{q}$. In fact, it has only interacted with $\mathbf{a}$, and then only at a time before $\mathbf{a}$ had any interaction with $\mathbf{q}$!

(d) What Bob can know after Alice's measurement

As we see in the previous section, after Alice's Bell-basis measurement, Bob's qubit $\mathbf{q}$ is in a state which may differ 'slightly' from $|\alpha\rangle$, depending on the outcome of that measurement:

- If $(\mathbf{x}, \mathbf{z}) = (0, 0)$, then $\mathbf{b}$ is in the same state $|\alpha_{00}\rangle := a_0|0\rangle + a_1|1\rangle = |\alpha\rangle$ that $\mathbf{q}$ was before the Bell-basis measurement.
- If $(\mathbf{x}, \mathbf{z}) = (0, 1)$, then $\mathbf{b}$ is in the state $|\alpha_{01}\rangle := a_0|0\rangle - a_1|1\rangle = Z|\alpha\rangle$.
- If $(\mathbf{x}, \mathbf{z}) = (1, 0)$, then $\mathbf{b}$ is in the state $|\alpha_{10}\rangle := a_1|0\rangle + a_0|1\rangle = X|\alpha\rangle$.
- If $(\mathbf{x}, \mathbf{z}) = (1, 1)$, then $\mathbf{b}$ is in the state $|\alpha_{11}\rangle := a_1|0\rangle - a_0|1\rangle = ZX|\alpha\rangle$.

In each case, the state of $\mathbf{b}$ is $|\beta\rangle = |\alpha_{\mathbf{z}\mathbf{x}}\rangle = Z^{\mathbf{z}}X^{\mathbf{x}}|\alpha\rangle$, so all Bob has to do is apply the correct single-qubit transformation to get Alice's state $|\alpha\rangle$.

There is a catch, however: it is Alice, and not Bob, who knows the values of $\mathbf{x}$ and $\mathbf{z}$. In fact, until Alice tells Bob that she has performed her measurement, there is no way for Bob to know that it has happened. This is despite the fact that the state of $\mathbf{b}$ has 'collapsed'. The only information that Bob can get from $\mathbf{b}$, is by performing a measurement in some

basis $|u\rangle, |v\rangle$, as before. For each outcome $(\mathbf{x}, \mathbf{z})$ of Alice's measurement, the probability of Bob measuring $|u\rangle$ is given by:

$$\Pr[u | (\mathbf{x}, \mathbf{z})] = \langle \alpha_{\mathbf{z}\mathbf{z}} | u \rangle \langle u | \alpha_{\mathbf{x}\mathbf{z}} \rangle = \langle \alpha | X^{\mathbf{x}} Z^{\mathbf{z}} | u \rangle \langle u | Z^{\mathbf{z}} X^{\mathbf{x}} | \alpha \rangle \quad (4.7)$$

. However, we are interested in the probability of the outcome: not for one particular outcome of Alice's measurement, but independently of Alice's measurement. We then compute $\Pr[u]$, using conditional probabilities:

$$\begin{aligned} \Pr[u] &= \Pr[u | (0, 0)] \cdot \Pr[(0, 0)] + \Pr[u | (0, 1)] \cdot \Pr[(0, 1)] \\ &\quad + \Pr[u | (1, 0)] \cdot \Pr[(1, 0)] + \Pr[u | (1, 1)] \cdot \Pr[(1, 1)] \\ &= \frac{1}{4} \langle \alpha | u \rangle \langle u | \alpha \rangle + \frac{1}{4} \langle \alpha | Z | u \rangle \langle u | Z | \alpha \rangle \\ &\quad + \frac{1}{4} \langle \alpha | X | u \rangle \langle u | X | \alpha \rangle + \frac{1}{4} \langle \alpha | XZ | u \rangle \langle u | ZX | \alpha \rangle \\ &= \langle \alpha | \left(\frac{1}{4} |u\rangle\langle u| + \frac{1}{4} Z |u\rangle\langle u| Z + \frac{1}{4} X |u\rangle\langle u| X + \frac{1}{4} XZ |u\rangle\langle u| ZX \right) | \alpha \rangle . \end{aligned} \quad (4.8)$$

To evaluate this, we consider the value of the operator in parentheses on the final line:

$$\begin{aligned} &\frac{1}{4} |u\rangle\langle u| + \frac{1}{4} Z |u\rangle\langle u| Z + \frac{1}{4} X |u\rangle\langle u| X + \frac{1}{4} XZ |u\rangle\langle u| ZX \\ &= \frac{1}{4} \begin{bmatrix} u_0^* u_0 & u_0^* u_1 \\ u_1^* u_0 & u_1^* u_1 \end{bmatrix} + \frac{1}{4} Z \begin{bmatrix} u_0^* u_0 & u_0^* u_1 \\ u_1^* u_0 & u_1^* u_1 \end{bmatrix} Z + \frac{1}{4} X \begin{bmatrix} u_0^* u_0 & u_0^* u_1 \\ u_1^* u_0 & u_1^* u_1 \end{bmatrix} X + \frac{1}{4} XZ \begin{bmatrix} u_0^* u_0 & u_0^* u_1 \\ u_1^* u_0 & u_1^* u_1 \end{bmatrix} ZX \\ &= \frac{1}{4} \begin{bmatrix} u_0^* u_0 & u_0^* u_1 \\ u_1^* u_0 & u_1^* u_1 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} u_0^* u_0 & -u_0^* u_1 \\ -u_1^* u_0 & u_1^* u_1 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} u_1^* u_1 & u_1^* u_0 \\ u_0^* u_1 & u_0^* u_0 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} u_1^* u_1 & -u_1^* u_0 \\ -u_0^* u_1 & u_0^* u_0 \end{bmatrix} \\ &= \frac{1}{4} \begin{bmatrix} 2u_0^* u_0 + 2u_1^* u_1 & 0 \\ 0 & 2u_1^* u_1 + 2u_0^* u_0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \frac{1}{2} I . \end{aligned} \quad (4.9)$$

However, this means that the probability that Bob measures $|u\rangle$ immediately after Alice's Bell-basis measurement, is:

$$\Pr[u] = \langle \alpha | \left(\frac{1}{2} I \right) | \alpha \rangle = \frac{1}{2} \langle \alpha | \alpha \rangle = \frac{1}{2}. \quad (4.10)$$

This doesn't depend in any way on $|\alpha\rangle$ or $|u\rangle$ and is exactly the same as the probability that Bob measures $|u\rangle$ before Alice's measurement. In particular, there is no way for Bob to know whether Alice has measured her qubits, based on observations that he can make on his qubit. If he believes that Alice has measured her qubits, he can theorise that $\mathbf{b}$ is in principle in one of the states $|\alpha_{\mathbf{z}\mathbf{z}}\rangle$ for some $\mathbf{x}$ and $\mathbf{z}$. But he cannot tell which, and neither can he tell whether the measurement actually has happened.

Of course, the next thing that Alice does is to send the values of $(\mathbf{x}, \mathbf{z})$ to Bob. Once Bob knows the value of $\mathbf{x}$ and $\mathbf{z}$, he knows that his state is the specific state $|\alpha_{\mathbf{z}\mathbf{z}}\rangle = Z^{\mathbf{z}} X^{\mathbf{x}} |\alpha\rangle$. He can then recover the state $|\alpha\rangle$ that Alice wished to send, by applying the transformations $|\alpha_{\mathbf{z}\mathbf{z}}\rangle \xrightarrow{X^{\mathbf{x}} b} |\alpha_{0\mathbf{z}}\rangle \xrightarrow{Z^{\mathbf{z}} b} |\alpha_{00}\rangle$.

(e) Observations

In quantum teleportation, Alice manages to send a single-qubit state $|\alpha\rangle$ to Bob, without having to know anything about that state and without having to send any quantum system to Bob. This is assuming that they have arranged to share a particular kind of shared entangled state beforehand. All she needs to do is perform some measurements on qubits under her control and then send the measurement results to Bob, without any further interaction between her qubits and Bob's.

This procedure does not allow Alice to transmit information to Bob faster than the speed of light, however. Bob is unable to do anything particularly useful with his qubit until he receives the message from Alice about her measurement results – a message which must travel at the speed of light, or slower.

There are a few ways to try to provide an intuition about why quantum teleportation works, which amounts to an interpretation of the mathematics just explained. In effect, the entanglement provides a form of shared quantum correlation that allows Alice to encode the continuum of possible single-qubit states in two bits of information.

Operationally, this is all similar to how you can securely send a secret message using a one-time pad (or ‘Vernam cipher’), only using quantum correlations instead of a shared source of random bits. While Bob's qubit **b** never interacts directly or indirectly with **q**, it has as strong a correlation as possible with Alice's qubit **a**. When Alice performs her measurement on **q** and **a**, the measurement she performs is **also** one measuring correlations between them. While initially they have no correlation, the nature of quantum measurement forces them to be correlated or anticorrelated in particular ways. As a side-effect, the correlations that **a** originally had with **b**, result in **b** having a state similar to the state that was originally in **q**. The state of **q** is then ‘transmitted’ through the quantum correlations between **a** and **b**, and the outcome of the Bell-state measurement.

This procedure is not in itself a form of computation. However, similar procedures can be used to realise two-qubit operations between distant qubits, using a process called ‘gate teleportation’. It is also possible to realise unitary transformations directly through measurements on large networks of entangled qubits, known as **measurement-based quantum computation** (or ‘one-way’ quantum computation). For this reason, teleportation may play an important role in practical quantum computation, even though on its own it transmits quantum states rather than transforming them.

4.2 Hadamard operations and phase estimation

Perhaps the most well-known, ‘purely’ quantum operation is the single-qubit Hadamard operation:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}. \quad (4.11)$$

An attractive feature of the Hadamard matrix is that it interchanges the computational basis on a single qubit (the states on the z -axis) and the $|+\rangle, |-\rangle$ basis (on the x -axis). But when applied to many qubits, it performs the important role of preparing **uniform superpositions**.

On n qubits, the **uniform superposition** $|u\rangle$ is the state in which every Boolean string occurs with the same amplitude:

$$|u\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle = \frac{1}{\sqrt{2^n}} |00 \cdots 00\rangle + \frac{1}{\sqrt{2^n}} |00 \cdots 01\rangle + \cdots + \frac{1}{\sqrt{2^n}} |11 \cdots 11\rangle . \quad (4.12)$$

The way that we may prepare $|u\rangle$, is by performing the operation H on each qubit of an n -qubit register $\mathbf{q}$, initialised to $|00 \cdots 0\rangle$. We may describe the effect of this as follows:

$$\begin{aligned} |0, 0, \dots, 0\rangle &\xrightarrow{H \mathbf{q}} |+, +, \dots, +\rangle \\ &= \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) \otimes \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) \otimes \cdots \otimes \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) \\ &= \left(\frac{1}{\sqrt{2}} \sum_{x_1 \in \{0,1\}} |x_1\rangle \right) \otimes \left(\frac{1}{\sqrt{2}} \sum_{x_2 \in \{0,1\}} |x_2\rangle \right) \otimes \cdots \otimes \left(\frac{1}{\sqrt{2}} \sum_{x_n \in \{0,1\}} |x_n\rangle \right) \\ &= \left(\frac{1}{\sqrt{2}} \right)^n \sum_{x_1, x_2, \dots, x_n \in \{0,1\}} |x_1\rangle \otimes |x_2\rangle \otimes \cdots \otimes |x_n\rangle \\ &= \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x_1, x_2, \dots, x_n\rangle = |u\rangle . \end{aligned} \quad (4.13)$$

4.2.1 Massive parallelism: How not to use a quantum computer

The uniform superposition state on n qubits plays an important role in many quantum algorithms. The role of the uniform superposition state is **often** described as enabling computation on all possible bitstrings at once. However, as we mentioned in Week 3, this description is misleading. For instance: consider a unitary transformation U_f on $n+1$ qubits, which realises a transformation:

$$|x, t\rangle \xrightarrow{U_f} |x, t \oplus f(x)\rangle . \quad (4.14)$$

This could be realised by a QAAL subroutine `U.f`, by translating a reversible procedure to compute $(x, t) \xrightarrow{P_f} (x, t \oplus f(x))$ into `X`, `CX`, and `CCX` operations. If we apply this transformation to a quantum register $\mathbf{q}$ in the uniform superposition $|u\rangle$ and to a target qubit $\mathbf{t}$ in the state $|0\rangle$, the resulting transformation is:

$$|u, 0\rangle_{\mathbf{q}, \mathbf{t}} = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x, 0\rangle_{\mathbf{q}, \mathbf{t}} \xrightarrow{U.f \mathbf{q}, \mathbf{t}} \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x, f(x)\rangle_{\mathbf{q}, \mathbf{t}} . \quad (4.15)$$

This is a superposition of all possible values of $x \in \{0, 1\}^n$ in $\mathbf{q}$ and the value of $f(x)$ in $\mathbf{t}$. But if we were to perform a measurement of these qubits in the computational basis, all

we would obtain is some (uniformly) random value of $x \in \{0, 1\}^n$, and $f(x)$ for that value of x , which we could do just as easily without a quantum computer. (Even if we were to measure just $\mathfrak{t}$, the probability with which we obtain $|0\rangle$ or $|1\rangle$ would simply depend on the number of $x \in \{0, 1\}^n$ for which $f(x) = 0$ or $f(x) = 1$. That is, if the probabilities of the outcomes for $\mathfrak{t}$ don't change just because we aren't measuring $\mathfrak{q}$ at the same time).

4.2.2 Introducing phase kicks

The main point of using a quantum computer is to do a computation differently from how you would perform it using a classical computer. This often involves subroutines which **are** similar to what a classical computer would do. However, generally they are not used in a way that could be understood straightforwardly in classical terms. The simplest and, in a sense, the most widespread example of this concept, is the **phase kick**.

We may demonstrate phase kicks most simply using **CX** operations. This involves using the **CX** operation:

$$|x, t\rangle \xrightarrow{\text{CX}} |x, t \oplus x\rangle , \quad (4.16)$$

to modify the state of the first, ‘control’ qubit, depending on the state of the second, ‘target’ qubit. For this to be even possible, neither the control nor the target qubit can be in a computational basis state. Instead, we may consider what happens when these two qubits are in the $|\pm\rangle$ basis.

- If the ‘target’ qubit is in the $|+\rangle$ state, the state of the ‘control’ qubit is left unaffected (in fact, regardless of what state $|\alpha\rangle = a_0|0\rangle + a_1|1\rangle$ it is in) by the **CX** operation:

$$\begin{aligned} |\alpha\rangle \otimes |+\rangle &= a_0|0\rangle \otimes \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) + a_1|1\rangle \otimes \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \\ &\xrightarrow{\text{CX}} a_0|0\rangle \otimes \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) + a_1|1\rangle \otimes \left(\frac{1}{\sqrt{2}}|1\rangle + \frac{1}{\sqrt{2}}|0\rangle\right) \\ &= |\alpha\rangle \otimes |+\rangle . \end{aligned} \quad (4.17)$$

- If the ‘target’ qubit is in the $|-\rangle$ state, and the ‘control’ is in the state $|+\rangle$, then **CX** changes the state of the control to the state $|-\rangle$, and vice versa. For instance, we have:

$$\begin{aligned} |+\rangle \otimes |-\rangle &= \frac{1}{2}|0, 0\rangle - \frac{1}{2}|0, 1\rangle + \frac{1}{2}|1, 0\rangle - \frac{1}{2}|1, 1\rangle \\ &\xrightarrow{\text{CX}} \frac{1}{2}|0, 0\rangle - \frac{1}{2}|0, 1\rangle - \frac{1}{2}|1, 1\rangle + \frac{1}{2}|1, 0\rangle \\ &= \frac{1}{\sqrt{2}}|0\rangle \otimes \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right) - \frac{1}{\sqrt{2}}|1\rangle \otimes \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right) \\ &= |-\rangle \otimes |-\rangle . \end{aligned} \quad (4.18)$$

And because **CX** is self-inverse, we also have $|-\rangle \otimes |-\rangle \xrightarrow{\text{CX}} |+\rangle \otimes |+\rangle$.

While the **CX** operation might have **appeared** to act asymmetrically on the two qubits involved, this was just an artefact of our emphasis on the computational basis. We’re not

talking about Newtonian physics here, but the Third Law of Motion (‘for every action, there is an equal and opposite reaction’) still broadly applies. The ‘control’ and the ‘target’ of the **CX** operation in fact have an equally important influence on **each others’** states.

4.2.3 Phase kicks and phase estimation

We may apply this principle regarding the **CX** operation specifically, to a so-called ‘toy’ problem, known as the ‘Bernstein–Vazirani problem’.

(a) Oracle problems

The Bernstein–Vazirani problem uses a setting which is common in quantum computation: the setting of **oracle problems**. In these problems, we aren’t given an input in the form of a binary string, but rather in the form of a **black-box subroutine**: an ‘oracle’, in the jargon of theoretical computer science. These are usually presented as a reversible function **0.f** which computes $(x, t) \mapsto (x, t \oplus f(x))$ on a classical computer. We will upgrade this to a unitary transformation $|x, t\rangle \mapsto |x, t \oplus f(x)\rangle$ when considering quantum algorithms. We are not told how **0.f** is implemented, but we may be told some limited amount of information about the function f . For example, we may be told that it is either constant (only evaluating to 0 or 1 for all inputs) or ‘balanced’ (evaluating to 0 or 1 for an equal number of inputs). The aim is then to determine some feature of the function f . This might be to determine whether or not f is constant, or to determine whether there is some input x such that $f(x) = 1$.

Of course, classical computers are not restricted to reversible procedures and we can’t obtain a unitary transformation from an unknown reversible subroutine either. But any function which could be computed by an irreversible black-box procedure, could also be computed by a reversible black-box procedure, or a unitary one. It is also worth noting that we very rarely have the ability to compute a function without having the ability to learn **something** about its implementation (if only its machine code). The point is that there are some problems which you can solve efficiently in this setting, without knowing how the function is implemented. Also, that the functions for which you can do this will depend on whether or not you have a quantum computer to help.

(b) The Bernstein–Vazirani problem

The Bernstein–Vazirani problem is a simple example of an oracle problem. In it, we are given an oracle **0.f** with the promise that it computes a function $\mathbf{0.f} : \{0, 1\}^{n+1} \rightarrow \{0, 1\}^{n+1}$ of the form $\mathbf{0.f}(x, t) = (x, t \oplus f(x))$, where:

$$f(x) = s \cdot x = s_1 x_1 \oplus s_2 x_2 \oplus \cdots \oplus s_n x_n, \quad (4.19)$$

for some bitstring $s \in \{0, 1\}^n$. (Here, $s \cdot x$ represents the ‘dot product’ on binary strings, where the result is computed using the ‘XOR’ or ‘addition mod 2’ operation $\oplus$, rather than integer addition $+$). The aim is to find the value of s , using the oracle subroutine

`0_f` as few times as possible.

This is a so-called ‘toy’ problem, in part because this is a problem which we can easily solve with a classical reversible operation, calling the oracle subroutine `0_f` a total of n times. The approach is simple, simply evaluate:

$$\begin{aligned}
0_f(1000 \cdots 00, 0) &= (1000 \cdots 00, s_1), \\
0_f(0100 \cdots 00, 0) &= (0100 \cdots 00, s_2), \\
0_f(0010 \cdots 00, 0) &= (0010 \cdots 00, s_3), \\
&\vdots \\
0_f(0000 \cdots 01, 0) &= (0000 \cdots 01, s_n),
\end{aligned} \tag{4.20}$$

effectively looking up the bits of $s \in \{0, 1\}^n$ one at a time. It is possible to show that this is the best you can do, which may be unsurprising. That is, if you use `0_f` to learn one bit at a time, you need to use it n times to learn n bits of information. However, as we see in the next section, we can in fact learn the entire string $s \in \{0, 1\}^n$ in a single query, if we can apply `0_f` as a unitary transformation $|x, t\rangle \xrightarrow{0_f} |x, t \oplus f(x)\rangle$.

(c) The Bernstein–Vazirani algorithm

While we don’t know how `0_f` is implemented, we do know a way that it **could** be implemented: a way that we could model its internal dynamics, if you like. (Note also that if `0_f` is actually implemented a different way, this has no impact on any algorithm we might use to solve the problem). The way that `0_f` **could** be implemented, is by some procedure as demonstrated in Figure 4.4. In this hypothetical implementation, ‘`ClassicallyCtrlCX`’ is a procedure which performs a `CX` operation or not, depending on whether a classical ‘control’ bit is set to 1. We think of `s` as a variable hidden in the same place that the implementation of `0_f` is hidden, and where `s[j]` determines whether a `CX` operation occurs with `q[j]` as the control and `t` as the target. As we’ve seen, though, we can use that `CX` operation as a way of feeding information from the target back to the control. We can do this by setting the initial state of the **control** to $|+\rangle$ and the state of the target to $|-\rangle$.

We can solve the Bernstein–Vazirani problem, using the algorithm presented in Figure 4.5. This algorithm simply applies the phase-kick phenomenon to each of the qubits `q[j]` which are controls, in our hypothetical implementation of `0_f`. For each `CX` operation which is actually performed in `0_f`, by means of a `ClassicalCtrlCX` operation, a qubit `q[j]` is flipped from $|+\rangle$ to $|-\rangle$. (The locations of those qubits are precisely those for which `s[j] = 1`). Having done this, we can map the qubits `q[j]` from the $|+\rangle, |-\rangle$ basis to the $|0\rangle, |1\rangle$ basis. The positions in which `q[j]` has the state $|1\rangle$ will then exactly coincide with the positions in which `s[j] = 1`. We can then simply measure `q` in the computational basis to find the value of `s`.

In the preceding explanation, we solve the Bernstein–Vazirani problem by preparing `q` in the uniform superposition, and then applying `0_f` to it. However, we do so not to

Figure 4.4: A possible implementation of the Bernstein–Vazirani oracle. Though the oracle `O.f` in the Bernstein–Vazirani problem does not have a specified implementation, it is possible to describe hypothetical implementations which can help us to solve the problem. One such implementation follows, involving a bit register `s` which is used in the realisation of the effect of `O.f`, by controlling whether CX operations are performed between control qubits `q[j]` and the target qubit `t`.

```

register s: n bits
# describe 's' as a globally visible register of some length n

# ... some operations to initialise the value of s ...

define O.f q, t :
    # a hypothetical implementation of the Bernstein--Vazirani
    oracle
    operand register q: n qubits          # for some integer n
    !integer j := n                       # a loop counter
    @loop:
        @jump_if_zero ctr, @done          # test for end of loop
        !set j := j - 1                   # decrement counter
        ClassicalCtrlCX s[j], q[j], t    # do 'CX q[j], t' if s[j] = 1
        !jump @loop                       # go to next iteration
    @done:
    stop

```

Figure 4.5: The Bernstein–Vazirani algorithm. This is a quantum procedure to solve the Bernstein–Vazirani problem. Here, n is some integer which defines the length of some bit-register `s`, which in turn is involved in the implementation of the subroutine `O.f`.

```

define BernsteinVazirani_Solver r
    # Solve the Bernstein--Vazirani problem, and produce the result
    # in a bit register r
    operand register r: n bits # for n the length of s
    register q: n qubits      # local quantum memory, to
    qubit t                   # solve the problem

    MzReset q                # reset all qubits in q to state |0>
    MzReset t                 # reset t to state |0>
    H q                       # map every qubit of q to the state |+>
    Ry(-pi/2) t               # map t into the state |->
    O.f q, t                  # use the Bernstein-Vazirani oracle on q and t
    H q                       # map every qubit of q to the states |0>, |1>
    Mz q -> r                 # extract the states of the qubits of q
    stop

```

evaluate `0.f` ‘massively in parallel’, but to probe some of the hidden dynamics of `0.f`. Even if we knew the implementation of `0.f`, it could be written in an obfuscated and unhelpful way. However, this does not matter so long as we know enough about `0.f` to be able to test its properties, like plucking a guitar string.

4.2.4 Measuring observables with eigenvalues ± 1

A different, somewhat more-complex application of phase kicks is in measuring observables. One way that we can interpret how phase kicks work for `CX` operators, is to consider the structure of the `CX` operator as in Equation (3.61):

$$\text{CX} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X. \quad (4.21)$$

If we observe that $|-\rangle$ is the -1 -eigenstate of X (and is a $+1$ -eigenstate of the identity operator I , as are all state-vectors), we can simplify the computation of Equation (4.18) by observing that:

$$\frac{1}{\sqrt{2}} |0, -\rangle + \frac{1}{\sqrt{2}} |1, -\rangle \xrightarrow{\text{CX}} \frac{1}{\sqrt{2}} |0, -\rangle - \frac{1}{\sqrt{2}} |1, -\rangle \quad (4.22)$$

simply because the control qubit (coherently) determines whether X gets applied to $|-\rangle$, where $|-\rangle$ is the -1 -eigenstate of X .

We may generalise this observation, to describe how we can implement a measurement of any n -qubit observable M with eigenvalues ± 1 , provided that we can write a `QAAL` program to realise M as a unitary transformation. (We’ve described the fact that we can **in principle** perform an M -observable measurement: the following provides an actual approach to do so). As M is a normal matrix with eigenvalues ± 1 , it **is** a unitary transformation. The only question is how to realise it as a sequence of operations.

Suppose that we have some subroutine `M`, expressed in terms of `Rx`, `Rz` and `CX` operations (for example), which realises $M : \mathbb{C}^{2^n} \rightarrow \mathbb{C}^{2^n}$ as a transformation of n -qubit states. We can then realise a subroutine `CM`, with an extra ‘control’ bit operand `c`, by:

- replacing every operation ‘`Rx(alpha) q`’ with ‘`CRx(alpha) c, q`’
- replacing every operation ‘`Rz(alpha) q`’ with ‘`CRz(alpha) c, q`’ and
- replacing every operation ‘`CX p, q`’ with ‘`CCX c, p, q`’.

This then realises the unitary transformation:

$$\text{CM} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes M. \quad (4.23)$$

What then happens when we apply this operation to the state $|+\rangle \otimes |\psi\rangle$, for an eigenstate (whether a $+1$ -eigenstate or -1 -eigenstate) of M , is that the control is either left unchanged or flipped to $|-\rangle$, depending on the eigenvalue a of $|\psi\rangle$:

$$\frac{1}{\sqrt{2}} |0, \phi\rangle + \frac{1}{\sqrt{2}} |1, \phi\rangle \xrightarrow{\text{CM}} \frac{1}{\sqrt{2}} |0, \phi\rangle + \frac{1}{\sqrt{2}} \cdot a |1, \phi\rangle = \begin{cases} |+, \phi\rangle & \text{if } a = +1, \\ |-, \phi\rangle & \text{if } a = -1. \end{cases} \quad (4.24)$$

If we measure the ‘control’ in the x -basis, we then obtain information about whether the eigenvalue of $|\phi\rangle$ is $+1$ or -1 .

If we perform this procedure on a state $|\psi\rangle$ that is not an eigenvector of M , we can decompose $|\psi\rangle = c_+ |\phi_+\rangle + c_- |\phi_-\rangle$ as a linear combination of states $|\phi_\pm\rangle$ which are ± 1 -eigenvectors of M . The result will then be either the state $|+\rangle \otimes |\phi_+\rangle$ (occurring with probability $|c_+|^2$) or the state $|-\rangle \otimes |\phi_+\rangle$ (occurring with probability $|c_-|^2$). That is to say: we realise a measurement of the observable M , using the control qubit of the CM operator to register the distinction between the eigenvalues ± 1 of the observable M .

4.2.5 Phase estimation

We can generalise the preceding material, to describe procedures to ‘measure’ other unitary operators U – even if their eigenvalues are not ± 1 . Such operators U are not Hermitian, so are not ‘observables’ in the traditional sense. However, what we can do is attempt to realise a procedure that will **(a)** collapse an input state $|\psi\rangle$ state onto an eigenvector $|\phi_a\rangle$ of U , and **(b)** obtain information characterising the eigenvalue a of the state $|\phi_a\rangle$.

The scalars $a = \pm 1$ are both ‘second roots of unity’, in that in both cases we have $a^2 = +1$. In this case, we have $a = e^{i\theta}$, for either $\theta = 0$ or $\theta = \pi$, which have in common that 2θ is a multiple of 2π (an integer number of whole rotations). More generally, if $\theta = \frac{2\pi k}{d}$ for some integers k and d , then $a = e^{i\theta}$ is a d^{th} root of unity, in that $a^d = +1$ – again, because $d \cdot \theta$ is an integer multiple of 2π).

The way that we generalise the phase kick to obtain information for an eigenvalue $a = e^{i\theta}$ for a more-general unitary transformation, is to attempt to obtain either the exact value of θ or a good approximation, expressed as a fraction of 2π . If $\theta = 2\pi \cdot \frac{n}{d}$, then obtaining a good approximation of $\frac{k}{d}$ will be enough for us to determine θ . This is a process known as **eigenvalue estimation** and is a key component of algorithms such as Shor’s algorithm.

In the next section, we will sketch some examples where eigenvalue estimation is more straightforward. Later, this concept will become essential for understanding Shor’s algorithm.

(a) Phase estimation for fourth roots of unity

Suppose that we have a QAAL subroutine U realising a unitary transformation on n qubits, realised using **CX**, **Rz** and **Rx** instructions. (We will also write that unitary transformation, by U). Further suppose that we happen to know that $U^4 = I$. It follows that all of the eigenvalues of U are **fourth** roots of unity, which is to say integer powers of $e^{i\pi/2} = i$: elements from the set $\{+1, +i, -1, -i\}$. (If also $U^2 \neq I$, we know that U doesn’t just have the eigenvalues ± 1 , so there must be some eigenstates with eigenvalue $+i$ or $-i$).

As before, we may use the implementation of U to implement a subroutine **CU**, in which we change every operation **CX**, **Rz** and **Rx** by their coherently controlled variants **CCX**, **CRz**

and **CRx**. We then have:

$$\mathbf{CU} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U \quad (4.25)$$

. Consider the effect of doing the phase-kick procedure, performing this operation on the state $|+\rangle \otimes |\phi_a\rangle$ for some eigenstate $|\phi_a\rangle$ of U , with eigenvalue $a = i^k \in \{+1, +i, -1, -i\}$. The resulting transformation is:

$$\begin{aligned} \frac{1}{\sqrt{2}} |0, \phi_a\rangle + \frac{1}{\sqrt{2}} |1, \phi_a\rangle &\xrightarrow{\mathbf{CU}} \frac{1}{\sqrt{2}} |0, \phi_a\rangle + \frac{i^k}{\sqrt{2}} |1, \phi_a\rangle \\ &= \begin{cases} |+, \phi_a\rangle & \text{if } a = +1, & |-, \phi_a\rangle & \text{if } a = -1, \\ |+i, \phi_a\rangle & \text{if } a = +i, & |-i, \phi_a\rangle & \text{if } a = -i. \end{cases} \end{aligned} \quad (4.26)$$

Again, we manage to arrange for the state of the ‘control’ qubit to enter a state which directly reflects the eigenvalue $a \in \{+1, +i, -1, -i\}$ of the state $|\phi_a\rangle$.

There is an important difference between this situation and the examples of phase kicks that we have seen already: the states $|+\rangle, |+i\rangle, |-\rangle, |-i\rangle$ are not all perfectly distinguishable from one another. (Note that $\langle + | +i \rangle = \frac{1}{\sqrt{2}} \langle 0 | +i \rangle + \frac{1}{\sqrt{2}} \langle 1 | +i \rangle = \frac{1+i}{2}$, for example, so that $|\langle + | +i \rangle|^2 = \frac{1}{2}$ rather than 0). So the preceding procedure alone cannot be used to measure the eigenvalue of a state $|\phi_a\rangle$.

Intuitively, the problem is that we cannot distinguish four possible outcomes with only one qubit of data. The natural solution is to try to find a way to do it with **two** qubits of data. To do this, it will be helpful to describe the transformation between a basis of states with coefficients related to powers of i , and the computational basis. Consider the following states, where we index the computational basis by $|0\rangle, |1\rangle, |2\rangle, |3\rangle \in \mathbb{C}^4$:

$$\begin{aligned} |f_{+1}\rangle &= \frac{1}{2}(|0\rangle + |1\rangle + |2\rangle + |3\rangle), & |f_{-i}\rangle &= \frac{1}{2}(|0\rangle + i|1\rangle - |2\rangle - i|3\rangle), \\ |f_{-1}\rangle &= \frac{1}{2}(|0\rangle - |1\rangle + |2\rangle + |3\rangle), & |f_{+i}\rangle &= \frac{1}{2}(|0\rangle - i|1\rangle - |2\rangle + i|3\rangle). \end{aligned} \quad (4.27)$$

Like the states $|\pm 1\rangle$, the states $|f_{i^k}\rangle$ have amplitudes of the same magnitude for each computational basis state. Only, for $|f_{i^k}\rangle$, the relative phases are powers of i instead of powers of -1 . (To be precise, we have $\langle x | f_{i^k} \rangle = \frac{1}{2} i^{xk}$ for $x, k \in \{0, 1, 2, 3\}$). It is not difficult to show that these states are orthogonal, for instance:

$$\langle f_{+i} | f_{+1} \rangle = \frac{1}{4} + \frac{1}{4}i - \frac{1}{4} - \frac{1}{4}i = 0. \quad (4.28)$$

As an alternative to doing a phase kick with a single qubit, we can do phase kicks with the two qubits used to represent the states $|f_{i^k}\rangle$, to help distinguish eigenvalues of $\pm i$, from eigenvalues of ± 1 . We do this by:

1. initialising a ‘control’ register c of two qubits to the state $|u\rangle = |+, +\rangle = |f_{+1}\rangle$
2. using **CU** performing a phase kick between $c[0]$ and a ‘target’ register q on which we want to do the eigenvalue measurement

3. using **CU** two more times, to perform two phase kicks between $c[1]$ and the target register q .

Suppose that q is in an eigenstate $|\phi_a\rangle$ of U , where $a \in \{+1, +i, -1, -i\}$. The effect of the **CU** operations described in the preceding explanation is:

$$\begin{aligned}
|+, +, \phi_a\rangle &= \frac{1}{2} \left(|00, \phi_a\rangle + |01, \phi_a\rangle + |10, \phi_a\rangle + |11, \phi_a\rangle \right) \\
&\xrightarrow{\text{CU } c[0], q} \frac{1}{2} \left(|00, \phi_a\rangle + a |01, \phi_a\rangle + |10, \phi_a\rangle + a |11, \phi_a\rangle \right) \\
&\xrightarrow{\text{CU } c[1], q} \frac{1}{2} \left(|00, \phi_a\rangle + a |01, \phi_a\rangle + a |10, \phi_a\rangle + a^2 |11, \phi_a\rangle \right) \\
&\xrightarrow{\text{CU } c[1], q} \frac{1}{2} \left(|00, \phi_a\rangle + a |01, \phi_a\rangle + a^2 |10, \phi_a\rangle + a^3 |11, \phi_a\rangle \right) \\
&= \frac{1}{2} \left(|00\rangle + a |01\rangle + a^2 |10\rangle + a^3 |11\rangle \right) \otimes |\phi_a\rangle = |f_a\rangle \otimes |\phi_a\rangle. \tag{4.29}
\end{aligned}$$

The benefit of doing this, is that while we may not know what the eigenvectors $|\phi_a\rangle$ of U are, we **do** know precisely what the states $|f_a\rangle$ are, and we can use this information to describe a practical method to find the value of $a = i^k$.

In particular, because the states $|f_a\rangle$ are orthogonal, we can define a unitary transformation F , as follows:

$$F = |0\rangle\langle f_{+1}| + |1\rangle\langle f_{+i}| + |2\rangle\langle f_{-1}| + |3\rangle\langle f_{-i}| = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & +i \end{bmatrix}. \tag{4.30}$$

The action of this operation can be described by $F|f_{ik}\rangle = |k\rangle$ for $k \in \{0, 1, 2, 3\}$. The operation F is actually one of a family of related unitary operations, known as **quantum Fourier transforms**. (The operation F is more commonly written as F_4 , because the transformation that it realises on the state-vector coefficients is the discrete Fourier transform over the integers modulo 4). This operation may be realised by the procedure **QFT_2** demonstrated in Figure 4.6. Using the operator F , we can then extract the power of i corresponding to the eigenvector $|\phi_a\rangle$:

$$|f_{ik}\rangle \otimes |\phi_{ik}\rangle \xrightarrow{\text{QFT}_2 \text{ } c} |k\rangle \otimes |\phi_{ik}\rangle. \tag{4.31}$$

By performing z -measurements on c , we can then obtain the integer k . This procedure is summarised in Figure 4.7.

(b) Phase estimation more generally

In the preceding code, there is nothing special about using two qubits for phase estimation. We don't need more than two qubits to extract phase information for a unitary U whose eigenvalues are all fourth roots of unity. However, if U had different eigenvalues from these, we would need a different procedure to obtain reliable information about

Figure 4.6: The quantum Fourier transform on two qubits. This is a procedure to realise the operator $F_4 : \mathbb{C}^{2^2} \rightarrow \mathbb{C}^{2^2}$ described in Equation (4.30), which is the Fourier transform modulo 4, or the quantum Fourier transform on two qubits.

```

define QFT_2 q :
  # The quantum Fourier transform,  $F_4$ , on two qubits.
  # We interpret q as a quantum register storing 2-bit integers
  operand register q: 2 qubits

  H q[1]
  CRz(pi/2) q[0], q[1]
  H q[0]
  SWAP q[0], q[1]
  stop

```

Figure 4.7: Phase estimation for the operator U . This is a procedure to do phase estimation, in order to measure the eigenvalue of a state stored in a quantum register q , relative to an operator U known to have only the eigenvalues $\{+1, +i, -1, -i\}$. The result is stored in a bit-register r , used to represent an integer $\{0, 1, 2, 3\}$ representing an eigenvalue of $e^{ir\pi/2}$.

```

define PhaseEstimU q, r
  # a subroutine to measure eigenstates of U via phase kicks
  operand register q: n qubits    # the register to measure
  operand register r: 2 bits      # to store the outcome

  register c: 2 qubits            # control register for phase kicks
  MxReset c                      # reset c to state  $|00\rangle$ 
  H c                            # prep control register in  $|f_{+1}\rangle$  state

  CU c[0], q # one kick to accumulate possibly imaginary phase
  CU c[1], q # two kicks to allow us to distinguish between
  CU c[1], q # imaginary phases and real-valued phases

  F c                            # map the Fourier basis onto the computational basis
  Mz c -> r # measure the outcome and store in the register r
  stop

```

them.

Fortunately, we can do this simply by generalising the procedure just covered. For instance: if in fact U had eigenvalues which are all **eighth** roots of unity (all of the form ω^k for $\omega = e^{ik\pi/4}$), then we could use three qubits for the control register c and perform a phase kick **four** ($= 2^2$) times, with the instruction ‘ $CU \ c[2], q$ ’. For an eigenstate $|\phi_a\rangle$ for $a = \omega^k$ for $k \in \{0, 1, \dots, 7\}$, this would yield:

$$\begin{aligned} |+, +, +, \phi_a\rangle &= \frac{1}{2\sqrt{2}} \left(|000, \phi_a\rangle + |001, \phi_a\rangle + |010, \phi_a\rangle + \dots + |111, \phi_a\rangle \right) \\ &\xrightarrow{\dots} \frac{1}{2\sqrt{2}} \left(|000, \phi_a\rangle + a |001, \phi_a\rangle + a^2 |010, \phi_a\rangle + \dots + a^7 |111, \phi_a\rangle \right) \\ &= \frac{1}{2\sqrt{2}} \left(|0\rangle + a |1\rangle + a^2 |2\rangle + \dots + a^7 |7\rangle \right) \otimes |\phi_a\rangle . \end{aligned} \quad (4.32)$$

Define the states $|f_{\omega^k}\rangle = \frac{1}{2\sqrt{2}} (|0\rangle + \omega^k |1\rangle + \omega^{2k} |2\rangle + \dots + \omega^{-k} |7\rangle)$: the control register is then in the state $|f_{\omega^k}\rangle$. We may then define the quantum Fourier transform over the integers mod 8, by the unitary matrix:

$$F_8 = |0\rangle\langle f_{\omega^0}| + |1\rangle\langle f_{\omega^1}| + \dots + |7\rangle\langle f_{\omega^{-1}}| = \frac{1}{2\sqrt{2}} \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \dots & \omega^{-1} \\ 1 & \omega^2 & \omega^4 & \dots & \omega^{-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{-1} & \omega^{-2} & \dots & \omega \end{bmatrix} . \quad (4.33)$$

This can be realised by a QAAL procedure similar to QFT_2, and it can be used to transform the state $|f_{\omega^k}\rangle$ of the control register to an integer $|k\rangle$ which we can measure in the computational basis.

We can generalise this still further. For a unitary operator U with a wider range of eigenvalues, we can extend this by using m qubits in the control register (where m is any integer). By performing the appropriate phase kicks, we can perform the transformation:

$$|+\rangle^{\otimes m} \otimes |\phi_a\rangle \xrightarrow{\dots} \frac{1}{\sqrt{2^m}} \left(|0\rangle + a |1\rangle + a^2 |2\rangle + \dots + a^{2^m-1} |2^m-1\rangle \right) \otimes |\phi_a\rangle , \quad (4.34)$$

where $|+\rangle^{\otimes m}$ means the tensor product of m copies of $|+\rangle$, where a is **any** complex unit and $|\phi_a\rangle$ is an a -eigenvector of some unitary operator U . Suppose that the eigenvalues of a are all of the form $a = e^{2\pi i t}$, $a' = e^{2\pi i t'}$, and so forth, for $0 \leq t, t' \dots < 1$. If $|t' - t|$ is larger than $1/2^{m-1}$ for all such eigenvalues, then the possible states of the control after these phase kicks are nearly orthogonal. By applying an appropriate version of the quantum Fourier transform to the control register and then measuring it, we would then (with high probability) obtain an integer $0 \leq k < 2^m$ representing an approximation $k/2^m \approx t$, with m bits of precision. (In this case, where the values of t are restricted to integer multiples of $1/2^m$, we can obtain exact results with certainty, as just explained, for fourth and eighth roots of unity).

We will revisit these ideas when we come to discuss Shor's algorithm. Most importantly, as long as we have the ability to describe a coherently controlled version $\mathbf{CU}$ of a unitary $\mathbf{U}$ and know of some restrictions on the range of its eigenvalues, we can realise a measurement which projects onto the eigenvectors of $\mathbf{U}$ (which might otherwise be difficult to realise).

Summary

We have finally begun to see some of the interesting aspects of what can be done with quantum information processing. The main observations are:

- entanglement can allow us to perform communication tasks which are not obvious from a classical perspective
- computing on superposition states means that information about eigenvalues related to the operation that we perform might be useful and important
- in particular, using information about eigenvalues can let us extract information about subroutines in a way which is not directly possible in a classical computation
- useful quantum computations do not approach things simply by computing functions 'massively in parallel', but must also use operations in non-obvious ways.